

**OPTIMIZING CUSTOMER SUPPORT IN BANKING THROUGH  
GENERATIVE ARTIFICIAL INTELLIGENCE: AN INTERNSHIP  
EXPERIENCE AT A BANK**

by

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## Declaration

I hereby declare that this thesis is my original work and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in the thesis.

This thesis has also not been submitted for any degree in any university previously.

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## Chapter 1

### Executive Summary

During my internship at the bank, I worked on two key projects to enhance customer service using Generative Artificial Intelligence (GenAI). The first project was developing a GenAI-powered chatbot to streamline customer interactions, using retrieval-augmented generation (RAG) techniques for accurate responses. The chatbot was designed with robust security and data privacy measures.

The second project involved creating an evaluation framework for the RAG-based chatbot using the RAGAS framework. This system automated performance assessments with a database of 800 test questions, providing insights for continuous improvement and benchmarking.

Throughout the internship, I gained valuable experience in advanced coding practices, GenAI library usage, MLOps infrastructure, and data privacy and security. My professional communication skills were also enhanced through collaboration with different stakeholders.

## Chapter 2

## Background

### 2.1 Company Background

The bank is based in Singapore, established with the aim of transforming banking services for everyday consumers and small businesses. The bank is determined to leverage advanced technology and data analytics to create a banking experience tailored to the needs of its customers, particularly those underserved by traditional financial institutions.

### 2.2 Project Background

In today's world, customers expect seamless and efficient support from the bank. Traditional agent-assisted channels like phone calls and emails can be time-consuming. Additionally, with the rapid growth of digital banking services, immediate access to information and assistance for various banking tasks has become essential.

To address these challenges, the bank has developed a chatbot powered by Generative Artificial Intelligence (GenAI) that utilizes bank publicly available FAQ articles as the source of data to generate responses. The technique retrieval-augmented generation (RAG) is applied, allowing the chatbot to answer user queries with bank internal data. This chatbot will handle customer inquiries, allowing Customer Support specialists to focus on more complex issues. The key objectives are to increase self-service rates, provide quick access to information, and enhance the overall efficiency of the customer support.

## 2.3 Generative Artificial Intelligence

Generative Artificial Intelligence (GenAI) is an innovative technology in natural language processing, capable of producing coherent and contextually relevant text. The core of GenAI is the Large language model (LLM), which stems from two key developments in deep learning and transformer architectures. Vaswani et al. (2017) introduced the transformer model, which applies a self-attention mechanism to efficiently capture relationships across words and phrases, allowing for the parallel processing of entire sentences and long-range dependencies [14]. Building on this foundation, Brown et al. (2020) presented a language model, GPT-3, which demonstrated exceptional few-shot learning capabilities, meaning it could perform tasks with minimal examples [2]. Together, these works underscore GenAI's potential for applications across domains, particularly in generating human-like text, understanding complex queries, and adapting to various linguistic tasks.

## 2.4 Risks Associated with Generative Artificial Intelligence in Bank

The integration of GenAI into bank chatbots introduces risks that require careful consideration. A primary concern is the phenomenon of "hallucination" where models generate responses that appear credible but are factually incorrect or nonsensical. In the context of bank chatbots, such misinformation can lead to incorrect financial guidance, misrepresentation of bank product details, potentially causing customers to make ill-informed decisions and exposing the bank to legal liability (Fui-Hoon Nah et al., 2023) [7]. Another concern is the 'black box' nature of GenAI models, which limits transparency and complicates banks' ability to interpret or explain specific chatbot responses or to identify potential errors (Dwivedi et al., 2023; Rudin et al., 2019) [4, 12]. Without traceable decision-making processes, customer trust may erode, and accountability becomes challenging.

Ethical and security concerns are also pertinent in GenAI-powered bank chatbots. The LLM is trained on a vast amount of internet data, which may include harmful or discriminatory text. This can impact the model's responses and potentially lead to biased treatment toward different races, genders, or socio-economic groups (Fui-Hoon

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Nah et al., 2023) [7]. For instance, a chatbot might inadvertently provide preferential information to certain groups, violating the bank guidelines of fairness. The use of chatbots in customer interactions involves handling sensitive personal data, which creates potential risks around data security and privacy, including the chance of accidental exposure or unauthorized access to personal information (Gupta et al., 2023). Additionally, GenAI-based chatbots are susceptible to security threats such as prompt injection attacks. Malicious actors may input crafted prompts to manipulate the chatbot into revealing sensitive information or performing unauthorized actions (Esmradi et al., 2023) [5].

## Chapter 3

### Project 1: Customer Service Chatbot

#### 3.1 Background

This project focuses on developing a customer service chatbot that addresses critical challenges, including security, "hallucination", and data handling. The methodology outlines a robust architecture that ensures data privacy and operational scalability, ultimately enhancing the customer experience in the banking sector.

#### 3.2 Market Solutions

The integration of GenAI into banking customer service has seen significant advancements across several financial institutions. These developments aim to enhance operational efficiency, improve customer experiences. The following examples illustrate how banks have employed GenAI technologies in their customer service chatbot solutions.

DBS Bank's implementation exemplifies a conventional approach through its Digibot platform, utilizing a rule-based system architecture. This system operates on predetermined response patterns and structured workflows, prioritizing operational reliability over conversational flexibility. The technical infrastructure incorporates essential security protocols, including two-factor authentication and end-to-end encryption for sensitive data transmission. Empirical observations of Digibot's functionality reveal its effectiveness in executing routine banking operations, such as loan application processing, transaction history retrieval, and account information access. However, the system's architectural limitations manifest in its restricted semantic understanding, relying primarily on keyword recognition rather than contextual comprehension. This design choice reflects a deliberate prioritization



### CHAPTER 3. PROJECT 1: CUSTOMER SERVICE CHATBOT

of transactional accuracy over conversational sophistication, resulting in high task completion reliability within defined parameters (DBS, n.d.) [3].

Bank of America’s Erica system represents an intermediate technological approach, incorporating natural language processing and machine learning capabilities while deliberately excluding GenAI components. The system architecture employs traditional machine learning models for query interpretation and response selection from predefined options. This methodology demonstrates a measured approach to AI implementation, emphasizing controlled responses while maintaining conversational fluency. Notable technical features include the integration of continuous learning mechanisms, whereby user interactions contribute to systematic response refinement under human supervision. The system’s operational scope encompasses standard banking functions and proactive financial monitoring, including merchant refund tracking and duplicate charge detection. Security implementation follows standard banking protocols, with the system operating exclusively within the bank’s authenticated mobile application environment. Performance data indicates successful query resolution within defined operational parameters, though the deliberate exclusion of generative AI technology establishes clear boundaries in terms of response flexibility (Bank of America, n.d.) [1].

JPMorgan Chase’s IndexGPT represents the most technologically advanced implementation among the examined systems, specifically in the institutional banking segment. The system’s architecture integrates OpenAI’s GPT-4 large language model, demonstrating the practical application of generative AI in specialized financial services. Technical analysis reveals its primary function in thematic keyword generation for investment portfolio construction, operating within the bank’s proprietary Quest framework. The system demonstrates particular efficiency in processing complex investment themes, including artificial intelligence, cloud computing, and renewable energy sectors. It enhances operational efficiency in index construction processes, though the implementation maintains significant human oversight components. This hybrid approach of advanced AI capability and human supervision represents a methodological balance between technological innovation and operational risk management (J.P. Morgan, 2024) [8].

## 3.3 Methodology

This solution employs a multi-phase and multi-agent approach to streamline conversational interactions and address challenges related to security, "hallucination", and data handling.

### 1. Session Initialization

When a user initiates a conversation, a session is established. The system records the user's session ID to maintain continuity and context.

### 2. Data Sanitization

An external tool is utilized to automatically remove sensitive data and special characters from the user input, safeguarding privacy.

### 3. Language Detection

An external tool is used to detect the dominant language of the input. Unsupported languages trigger a standardized response, requesting the user to continue in a supported language.

### 4. Greeting Identification

The system checks if the input is a greeting. If confirmed, it provides a templated greeting response and resets the process for new user input.

### 5. History Retrieval

The user's session history is retrieved from the database, preserving conversation context.

### 6. Security Verification

LLM Security Agent: Processes the sanitized input to detect prompt injections, off-topic queries, or disallowed content. If any are detected, the system informs the user of content limitations and halts further processing.

### 7. FAQ Retrieval and Response Generation

LLM Retrieval Agent: Retrieves relevant content from a vector database containing the bank's FAQs and knowledge base. Then it constructs an accurate, context-aware response based on the retrieved information.

### 8. Output Sanitization

An external service removes any PII from the generated response before it is sent to the user.

### 9. Session Logging

The conversation log, including user inputs and chatbot responses, is securely stored in the database for analysis and future reference.

### 10. Loop Continuation

The process iterates, allowing for seamless conversational flow until the session ends.

## 3.4 Results

The chatbot's performance was evaluated across multiple key metrics. In terms of latency, the system maintained an average time of 4.5 seconds per interaction, ensuring prompt responses to user queries. User satisfaction was assessed based on feedback, with 5% of users providing a thumbs-up response, indicating a positive reception overall, while 4% of users gave a thumbs-down, highlighting areas for potential improvement. Regarding security effectiveness, the chatbot demonstrated robust capabilities by successfully identifying and mitigating 100% of simulated security threats, including prompt injections and sensitive queries, during testing. The empirical testing was conducted over a two-month period with 2000 sessions, providing a solid sample for analysis. During this period, user interactions were logged, ensuring compliance with data privacy regulations. Additionally, a feedback mechanism was implemented, encouraging users to share their experiences, which provided valuable insights for continuous improvement of the system.

## 3.5 Discussion

The chatbot's multi-phase algorithm is designed for efficiency, accuracy, and data privacy. Each phase addresses specific needs to ensure reliable and resource-efficient operation. A key feature is a layered verification and security approach, embedding early checks for security, language, and context to mitigate risks and safeguard data.

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This modular design integrates various models and tools, enhancing robustness and protecting sensitive information.

For contextual accuracy, the chatbot uses vector-based retrieval aligned with FAQ content and a history summarization phase, ensuring consistent and relevant responses. Conversation logs in a database facilitate efficient retrieval, analysis, and model fine-tuning.

The chatbot is scalable and efficient, distributing tasks across specialized agents like GPT-A for security checks and GPT-B for retrieval. This reduces computational strain and costs while allowing independent optimization of components.

Despite its robustness, sequential processing causes noticeable latency. Future improvements will focus on concurrent phase execution, fine-tuning smaller LLMs, and a distributed check system. Running security checks and response generation concurrently, and using specialized, smaller models for specific tasks, will reduce latency and enhance efficiency.

## Chapter 4

### Project 2: Chatbot Evaluation System

#### 4.1 Background

The inherent limitations of LLM necessitate robust monitoring mechanisms for quality assurance. In the banking context, continuous monitoring becomes particularly crucial when the base LLM model is updated, new features are implemented, or prompt engineering modifications are made. Such changes require a comprehensive evaluation framework to assess behavioral variations from previous versions and ensure maintained compliance with security protocols and ethical standards. To address this need, the bank has developed an extensive test question database comprising 800 questions, curated through cross-functional collaboration with various business units.

However, the manual evaluation of chatbot responses presents significant operational challenges. A comprehensive assessment of 800 test questions typically requires more than two working days of human effort. Moreover, human evaluators often introduce subjective biases, particularly when assessing responses to open-ended questions. These challenges underscore the need for an automated evaluation system capable of objectively assessing chatbot responses against test cases while providing explainable results.

#### 4.2 Literature Review

Traditional evaluation metrics for natural language processing tasks have primarily relied on textual similarity measures. ROUGE (Lin, 2004) emphasizes recall by measuring n-gram and longest common subsequence overlaps, while BLEU (Papineni et al., 2002) focuses on precision with n-gram matches and brevity penalties [9, 11].

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Although these metrics have become standard in the field, their effectiveness in RAG contexts is limited, particularly when semantic flexibility and coherence are crucial considerations.

To address the limitations of n-gram-based metrics, BERTScore (Zhang et al., 2019) was introduced as a more semantically nuanced evaluation method [15]. By leveraging contextualized embeddings from pre-trained transformer models, BERTScore computes cosine similarity between generated and reference sentences, enabling better assessment of semantic relationships. However, its dependence on specific embedding models introduces potential biases and domain-specific limitations.

While these traditional metrics focus on measuring similarity between ground truth and generated answers, they may not fully capture the complexities of RAG systems, where the relationship between retrieved context, ground truth, and generated answers requires more sophisticated evaluation approaches. This prompted the development of evaluation frameworks based on LLM.

G-Eval (Liu, Yang, et al., 2023) implements a multi-dimensional evaluation approach, utilizing LLMs to score responses across various metrics such as factual accuracy, coherence, and relevance [10]. The framework allows for customization of evaluation criteria but requires careful prompt engineering and may face limitations in terms of model compatibility.

GPTScore (Fu et al., 2023) takes a different approach by using conditional probability of generating target text as an evaluation metric, enabling zero-shot assessment of generated text quality [6]. Meanwhile, RAGAS combines model-based scores with rule-based heuristics to evaluate RAG systems, employing LLMs to extract relevant clauses from ground truth, contexts, and answers for score computation.

Building upon these approaches, ARES (Saad-Falcon et al., 2023) represents a more comprehensive evaluation framework that incorporates synthetic data generation, LLM-based judgment, and human preference validation [13]. The system generates synthetic queries and answers from in-domain passages, trains LLM judges using these materials, and employs prediction-powered inference to estimate confidence intervals for RAG system quality. This methodology aims to achieve higher human alignment in evaluation outcomes.

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This progression of evaluation methods reflects the field’s movement from simple textual similarity metrics toward more sophisticated approaches that better capture the nuanced requirements of RAG system evaluation, particularly in terms of context relevance and answer quality.

### 4.3 Methodology

The implementation of our RAG model evaluation framework presents a systematic solution for assessing model performance through automated processes and a robust infrastructure. Central to this framework is a multi-stage evaluation pipeline, integrated with cloud-native technologies to ensure scalability and operational efficiency.

#### Initial Setup

The evaluation system for the RAG-based chatbot was constructed utilizing the RAGAS framework, which was selected for its robustness and adaptability. The setup process involved several key steps:

##### Question Base Creation

A comprehensive database of 800 test questions was developed, covering four critical domains: Ethics & Fairness, Operational Risk, Exclusion List Compliance, and Product Scenarios. The Ethics & Fairness domain includes questions designed to evaluate the chatbot’s adherence to ethical guidelines and the absence of demographic biases. Operational Risk scenarios assess compliance with regulatory requirements in banking operations and personal account management. Exclusion List Compliance tests the chatbot’s ability to appropriately decline responses to restricted topics such as financial advice. Product Scenarios focus on inquiries related to banking products and services, emphasizing accurate and helpful user guidance.

##### Ground Truth Creation

For each test question, a corresponding ground truth answer was established through a process involving initial generation, human review, refinement, and validation. Responses were initially generated using the existing version of the

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chatbot. Expert evaluators, including compliance officers and customer service managers, reviewed these responses for accuracy and compliance. Adjustments were made to ensure the ground truth answers met all regulatory and ethical requirements, followed by a secondary review to confirm their quality and appropriateness.

### Metric Selection

From the extensive suite of metrics offered by RAGAS, four core metrics were selected: Retrieval Scores and Generation Scores. Retrieval Scores include Context Recall, which evaluates the extent to which relevant information from the knowledge base is retrieved in response to a query, and Context Precision, which assesses whether all ground-truth relevant items are appropriately ranked higher. Generation Scores include Faithfulness, measuring the consistency between the generated response and the retrieved context, and Answer Correctness, evaluating the accuracy of the generated answer compared to the ground truth.

### Evaluation Pipeline

1. **Extraction of Test Questions:** The evaluation pipeline initiates with the extraction of test questions and their corresponding ground truths from our database.
2. **Processing through the Chatbot Interface:** These extracted questions are then processed via the chatbot API, which generates responses based on the input questions. Concurrently, the chatbot retrieves relevant contextual information to inform its responses.
3. **Performance Metrics Computation:** Using the RAGAS framework, the pipeline evaluates the quality of chatbot responses and context retrieval, computing key performance metrics. This analysis examines relationships between ground truths, generated responses, questions posed, and retrieved contexts.
4. **Post-Processing of Results:** Following the computation of metrics, a post-processing step aggregates the results based on question categories. This aggregation provides granular insights into model performance across distinct question categories, facilitating targeted analysis.



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5. **Storage of Evaluation Results:** Finally, all evaluation results are persistently stored in a cloud-based database. This storage facilitates further analysis and also enables historical tracking of performance over time.

### Infrastructure

The technical architecture supporting this evaluation framework adheres to modern cloud-native principles. Source code management is conducted via Git, ensuring version control and facilitating collaborative development. Data management leverages robust data warehousing capabilities, providing secure and scalable storage for evaluation data. Evaluation metrics computation is performed using cloud computing services, which guarantee efficient resource utilization and scalability. Pipeline orchestration is managed through a workflow management platform, coordinating the execution of evaluation tasks and maintaining consistent scheduling.

This comprehensive architecture not only ensures the reliability and reproducibility of evaluation results but also maintains flexibility for future enhancements. The modular design allows for the integration of new evaluation metrics and methodologies, adapting to evolving requirements in RAG model evaluation.

### 4.4 Use Cases

The evaluation system is scheduled to run on a weekly basis, allowing for continuous monitoring of the chatbot's performance metrics. This consistent oversight helps identify potential performance drift and triggers alerts if the chatbot's effectiveness declines. For example, in one monitoring cycle, the system detected a 30% decrease in the context recall and faithfulness metrics. Further investigation revealed changes in the underlying data distribution, prompting necessary adjustments to realign the chatbot's responses with the expected standards.

The evaluation system rigorously assesses the chatbot across a variety of question categories, including data privacy, product details, and company regulations. By covering these diverse question types, the evaluation framework is able to pinpoint specific areas where the chatbot may underperform. This analysis enables targeted enhancements, allowing us to address weaknesses in particular domains, such

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as improving response accuracy for regulatory inquiries or expanding contextual knowledge for product-specific questions.

The evaluation system provides a robust framework for systematically comparing different versions of the chatbot, making it invaluable for tracking and showcasing changes, whether they arise from prompt engineering adjustments, updates to the contextual retriever, or changes in the underlying model itself. This functionality is crucial to the continued development and refinement of chatbot. It allows us to assess the impact of each modification on performance metrics in a controlled and measurable way. For example, following a base model upgrade, the evaluation system highlighted an improvement in faithfulness, with the updated model more accurately mirroring the specific phrasing from reference FAQs. While this increased fidelity contributed to accuracy, it also limited response variability. Based on these insights, we adapted the prompt design to reduce strict adherence to FAQ phrasing, enhancing conversational flexibility without compromising accuracy. This iterative evaluation and adjustment process strengthens the chatbot’s overall performance, ensuring it continues to meet evolving requirements.

### 4.5 Discussion

The RAGAS framework was chosen for its comprehensive set of evaluation metrics, which assess multiple stages of the RAG pipeline—from contextual retrieval accuracy through to the quality of generated responses. This multi-layered evaluation enables precise identification of performance issues within specific stages, allowing for targeted and efficient improvements. Another significant advantage is that RAGAS metrics closely align with human judgment, providing automated evaluations that accurately reflect real-world quality assessments. The framework’s flexibility to incorporate custom LLMs and embedding models also ensures compliance with internal guidelines, as only pre-approved models are utilized. Additionally, RAGAS is widely supported with extensive documentation and community resources, as evidenced by its popularity and active engagement (e.g., a GitHub rating of 7.1k stars as of November 2024). RAGAS operates in a zero-shot mode, meaning it does not require fine-tuning or prompt adjustments to execute evaluations, though it remains customizable if specific adaptation is needed. This adaptability and ease of

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use make RAGAS an efficient and reliable choice for this project’s evaluation needs.

Despite its strengths, RAGAS presents several challenges that were systematically addressed in this project. First, RAGAS requires a predefined set of questions and corresponding ground-truth answers, necessitating initial manual work to create an accurate question base. To address this, a diverse set of questions was curated, covering areas such as product inquiries and security-related topics. The question base was enriched through stakeholder input across departments, real user queries, and synthetic questions generated by RAGAS’s own LLM, referencing commonly asked questions. Establishing ground-truth answers also required manual intervention. These answers were initially derived from responses generated by the existing chatbot and subsequently validated by human evaluators. To maintain accuracy, this set of ground truths is updated regularly as the chatbot’s performance improves, ensuring that the benchmark responses reflect the highest achievable quality. The extensive suite of RAGAS metrics, exceeding eight distinct measures, can be challenging to interpret comprehensively. To streamline this, four core metrics were selected based on their ability to capture essential aspects of performance: context retrieval accuracy, generation quality, and overall response accuracy. These metrics were chosen for their low inter-correlation, ensuring that they effectively distinguish high-quality responses from lower-quality ones. Stability tests confirmed that these metrics produced consistent score distributions across multiple evaluation runs, enhancing reliability. Finally, non-determinism in LLM-based evaluations can lead to variations in scoring. To address this, the RAGAS evaluations were run multiple times, with results averaged to mitigate randomness. Additionally, the temperature parameter, which influences the model’s response creativity, was set to zero for both the evaluator and the chatbot. This adjustment ensures a more deterministic response by favoring high-probability tokens, resulting in outputs that are consistent and focused.

## Chapter 5

### Internship Takeaway

During the internship, significant knowledge was gained across several key areas including advanced coding practices, GenAI library usage, MLOps infrastructure, data privacy and security, and professional communication skills. Contributing to GenAI solutions emphasized disciplined coding practices such as modularization, reusability, comprehensive documentation, and effective code commenting, which are essential for ensuring readability, maintainability, and scalability. Proficiency with various GenAI libraries like Lambda Index, LangChain, and OpenAI was necessary, with a focus on strategic library selection, meticulous version control, and using virtual environments to manage dependencies and ensure code portability.

The design and deployment of MLOps infrastructure on platforms like AWS, AWS Lambda, AWS SageMaker, and Airflow provided practical insights into architecting scalable and secure GenAI applications. This required understanding distributed workflow management, efficient resource allocation, and the deployment of model evaluation pipelines, particularly in high-stakes environments such as finance. A significant challenge encountered was the limited documentation for evolving GenAI libraries like RAGAS, which necessitated extensive troubleshooting and direct source code analysis. This experience highlighted the importance of source code literacy and independent problem-solving skills when adapting to innovative but less-documented technological frameworks.

In the context of financial services, stringent data privacy requirements necessitated a careful approach to handling customer personally identifiable information and internal bank data. The chatbot design incorporated privacy-preserving mechanisms like PII masking and extensive security checks within the evaluation system to ensure compliance with data security standards. This experience underscored the criticality of incorporating data privacy and security protocols into the architecture of GenAI applications to safeguard customer data, maintain regulatory compliance,

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and mitigate risks associated with AI-driven customer interactions.

Additionally, the internship emphasized the importance of cross-disciplinary communication skills. Effectively conveying complex AI methodologies to non-technical stakeholders, such as project managers and executive-level leaders, required strategic use of infographics, PowerPoint, and written summaries. Collaborative discussions with data science and engineering teams further refined technical communication skills, enhancing the efficacy of cross-functional teamwork. Research capabilities were strengthened through systematic exploration of best practices, industry literature, and available solutions to complex challenges within GenAI.

## Chapter 6

### Conclusion

The internship at the bank provided a comprehensive learning experience, particularly in the application of GenAI technologies to improve customer service in the banking industry. The development of a GenAI-powered chatbot demonstrated significant potential in transforming customer interactions, enhancing efficiency, and ensuring robust security measures. The systematic evaluation of the chatbot using the RAGAS framework highlighted the importance of continuous performance monitoring and iterative improvements.

Key takeaways from this experience include the critical role of modular and reusable coding practices, the strategic selection and management of GenAI libraries, and the integration of stringent data privacy and security protocols. Additionally, the internship underscored the importance of effective communication and collaboration across diverse teams to achieve project goals.

Overall, this internship not only contributed to the advancement of customer service solutions within the bank but also provided invaluable insights and skills that will be instrumental in my future career in GenAI and data science.

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